

Input File Description

Program: neb.x / NEB / Quantum ESPRESSO (version: 7.5)

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INTRODUCTION

Input data format: { } = optional, [] = it depends, | = or

All quantities whose dimensions are not explicitly specified are in RYDBERG ATOMIC UNITS

BEWARE: TABS, DOS <CR><LF> CHARACTERS ARE POTENTIAL SOURCES OF TROUBLE

neb.x DOES NOT READ FROM STANDARD INPUT !

There are two ways for running a calculation with neb.x:

- (1) specifying a file to parse with the `./neb.x -inp` or `./neb.x -input` command line option.
- (2) or specifying the number of copies of PWscf inputs with the `./neb.x -input_images`

For case (1) a file containing special KEYWORDS (aka SUPERCARDS) has to be written (see below). These KEYWORDS tell the parser which part of the file

contains the neb specifics and which part contains the energy/force engine input (at the moment only PW). After the parsing, different files are generated: neb.dat, with the neb specific variables, and a set of pw_*.in Pwscf input files, i.e., one for each input position. All options for a single SCF calculation apply.

The general structure of the file to be parsed is:

=====

BEGIN

BEGIN_PATH_INPUT

... neb specific namelists and cards

END_PATH_INPUT

BEGIN_ENGINE_INPUT

...pw specific namelists and cards

BEGIN_POSITIONS

FIRST_IMAGE

...pw ATOMIC_POSITIONS card

...pw TOTAL_CHARGE card (only for FCP)

INTERMEDIATE_IMAGE

...pw ATOMIC_POSITIONS card

...pw TOTAL_CHARGE card (only for FCP)

LAST_IMAGE

...pw ATOMIC_POSITIONS card

...pw TOTAL_CHARGE card (only for FCP)

END_POSITIONS

... other pw specific cards

END_ENGINE_INPUT

END

For case (2) neb.dat and all pw_1.in, pw_2.in ... should be already present.

Structure of the NEB-only input data (file neb.dat):

=====

&PATH

...

/

[**CLIMBING_IMAGES**

list of images, separated by a comma]

BEGIN

Syntax of this supercard is the following:

BEGIN

... content of the supercard here ...

END

and the content is:

BEGIN_PATH_INPUT

Syntax of this supercard is the following:

BEGIN_PATH_INPUT

... content of the supercard here ...

END_PATH_INPUT

and the content is:

Namelist: **&PATH**

string_method CHARACTER

Default: 'neb'

A string describing the task to be performed. Options are:

'neb' :

nudget-elastic-band

'smd' :
string-method-dynamics

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restart_mode CHARACTER

Default: 'from_scratch'

Options are:

'from_scratch' :
from scratch

'restart' :
from previous interrupted run

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nstep_path INTEGER

Default: 1

number of ionic + electronic steps

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num_of_images INTEGER

Default: 0

Number of points used to discretize the path
(it must be larger than 3).

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opt_scheme CHARACTER

Default: 'quick-min'

Specify the type of optimization scheme:

'sd' :
steepest descent

'broyden' :
quasi-Newton Broyden's second method (suggested)

'broyden2' :
another variant of the quasi-Newton Broyden's
second method to be tested and compared with the
previous one.

'quick-min' :
an optimisation algorithm based on the
projected velocity Verlet scheme

'langevin' :
finite temperature langevin dynamics of the
string (smd only). It is used to compute the
average path and the free-energy profile.

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CI_scheme CHARACTER

Default: 'no-CI'

Specify the type of Climbing Image scheme:

'no-CI' :
climbing image is not used

'auto' :

original CI scheme. The image highest in energy does not feel the effect of springs and is allowed to climb along the path

'manual' :
images that have to climb are manually selected.
See also [CLIMBING_IMAGES](#) card

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first_last_opt LOGICAL

Default: .FALSE.

Also the first and the last configurations are optimized "on the fly" (these images do not feel the effect of the springs).

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minimum_image LOGICAL

Default: .FALSE.

Assume a "minimum image criterion" to build the path. If an atom moves by more than half the length of a crystal axis between one image and the next in the input (before interpolation), an appropriate periodic replica of that atom is chosen. Useful to avoid jumps in the initial reaction path.

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temp_req REAL

Default: 0.D0 Kelvin

Temperature used for the langevin dynamics of the string.

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ds REAL

Default: 1.D0

Optimisation step length (Hartree atomic units).
If [opt_scheme](#)=="broyden", ds is used as a guess for the diagonal part of the Jacobian matrix.

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k_max, k_min REAL

Default: 0.1D0 Hartree atomic units

Set them to use a Variable Elastic Constants scheme
elastic constants are in the range [k_min, k_max]
this is useful to rise the resolution around the saddle point.

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path_thr REAL

Default: 0.05D0 eV / Angstrom

The simulation stops when the error (the norm of the force orthogonal to the path in eV/A) is less than path_thr.

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use_masses LOGICAL

Default: .FALSE.

If. TRUE. the optimisation of the path is performed using mass-weighted coordinates. Useful together with quick-min optimization scheme, if some bonds are much stiffer than others. By assigning a larger (fictitious) mass to atoms with stiff bonds, one may use a longer time step "ds"

[\[Back to Top\]](#)**use_freezing** LOGICAL*Default:* .FALSE.

If .TRUE. the images are optimised according to their error:
only those images with an error larger than half of the largest
are optimised. The other images are kept frozen.

[\[Back to Top\]](#)**lfc** LOGICAL*Default:* .FALSE.*See:* [fcp_mu](#)

If .TRUE. perform a constant bias potential (constant-mu) calculation with
- ESM method (assume_isolated = 'esm' and esm_bc = 'bc2' or 'bc3' must be
set in SYSTEM namelist) or
- ESM-RISM method (assume_isolated = 'esm' and esm_bc = 'bc1' must be set
set in SYSTEM namelist, and trism = .TRUE. must be set
set in CONTROL namelist).

[fcp_mu](#) gives the target Fermi energy.
See the header of PW/src/fcp_module.f90 for documentation

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If [lfc](#) == .TRUE., gives the target Fermi energy [eV].
One can specify the total charge of the system for the first
and last image by giving [TOTAL_CHARGE](#) cards
so that the Fermi energy of these systems will be the target value,
otherwise [first_last_opt](#) should be .TRUE.
For the initial charge of intermediate images, the [TOTAL_CHARGE](#)
is linearly interpolated between the initial and the final ones
unless the [TOTAL_CHARGE](#) is given in the input file.

[\[Back to Top\]](#)**fcp_thr** REAL*Default:* 0.01D0 V*See:* [lfc](#)

The simulation stops when the error (the maximum of the force
acting on the FCP in V) is less than fcp_thr.

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Specify the type of optimization scheme for FCP:

'lm' :
Line-Minimization method.

'newton' :
Newton-Raphson method with diagonal hessian matrix.
Also, coupling with DIIS.

'coupled' :
Coupled method with ionic positions.
This is available only if [opt_scheme](#) == 'broyden',
or 'broyden2'.

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Card: CLIMBING_IMAGES

Optional card, needed only if [Cl_scheme](#) == 'manual', ignored otherwise !

Syntax:

CLIMBING_IMAGES
[index1, index2, ..., indexN](#)

Description of items:

index1, index2, ... indexN INTEGER
 index1, index2, ..., indexN are indices of the images to which the Climbing-Image procedure apply. If more than one image is specified they must be separated by a comma.

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END_PATH_INPUT**BEGIN_ENGINE_INPUT**

Syntax of this supercard is the following:

```
BEGIN_ENGINE_INPUT
... content of the supercard here ...
END_ENGINE_INPUT
```

and the content is:

Here comes the pw.x specific namelists and cards (see file: [INPUT_PW.html](#) or INPUT_PW.txt) with the exception of [ATOMIC_POSITIONS](#) cards, which are specified separately within the [BEGIN_POSITIONS](#)/END_POSITIONS supercard as described below.

So the input that follows here is of the following structure:

```
&CONTROL
...
/
&SYSTEM
...
/
&ELECTRONS
...
/
...
```

BEGIN_POSITIONS

Syntax of this supercard is the following:

```
BEGIN_POSITIONS
... content of the supercard here ...
END_POSITIONS
```

and the content is:

NB:
 Atomic positions for all the images are specified within the [BEGIN_POSITIONS](#) / END_POSITIONS supercard, where each instance of [ATOMIC_POSITIONS](#) card is prefixed either by [FIRST_IMAGE](#), [INTERMEDIATE_IMAGE](#), or [LAST_IMAGE](#) keywords.
 IF [Ifcp](#) == .TRUE., total charges for all images have to be specified with [TOTAL_CHARGE](#) cards.

Note that intermediate images are optional, i.e., there can be none or any number of [INTERMEDIATE_IMAGE](#) images.

FIRST_IMAGE

Syntax of this supercard is the following:

```
FIRST_IMAGE
... content of the supercard here ...
```

and the content is:

Card: ATOMIC_POSITIONS { **alat** | **bohr** | **angstrom** | **crystal** | **crystal_sg** }

For the description of ATOMIC_POSITIONS card see file: [INPUT_PW.html](#) or INPUT_PW.txt

Card: TOTAL_CHARGE

Optional card, needed only if **lfcf** == .TRUE.

Syntax:

```
TOTAL_CHARGE
tot_charge
```

Description of items:

tot_charge REAL tot_charge is the total charge of the system.

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INTERMEDIATE_IMAGE

Syntax of this supercard is the following:

```
INTERMEDIATE_IMAGE
... content of the supercard here ...
```

and the content is:

(**Remark:** There can be any number (including zero) of INTERMEDIATE_IMAGE supercards.)

Card: ATOMIC_POSITIONS { **alat** | **bohr** | **angstrom** | **crystal** | **crystal_sg** }

For the description of ATOMIC_POSITIONS card see file: [INPUT_PW.html](#) or INPUT_PW.txt

Card: TOTAL_CHARGE

Optional card, needed only if **lfcf** == .TRUE.

Syntax:

```
TOTAL_CHARGE
tot_charge
```

Description of items:

tot_charge REAL tot_charge is the total charge of the system.

LAST_IMAGE

Syntax of this supercard is the following:

LAST_IMAGE

... content of the supercard here ...

and the content is:

Card: ATOMIC_POSITIONS { **alat** | **bohr** | **angstrom** | **crystal** | **crystal_sg** }

For the description of ATOMIC_POSITIONS card see file: [INPUT_PW.html](#) or INPUT_PW.txt

Card: TOTAL_CHARGE

Optional card, needed only if [lfcp](#) == .TRUE.

Syntax:

TOTAL_CHARGE

[tot_charge](#)

Description of items:

tot_charge REAL

tot_charge is the total charge of the system.

END_POSITIONS

Here can follow other **pw** specific **cards** ...

END_ENGINE_INPUT

END